**Code No: 124CN** **R15**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B.Tech II Year II Semester Examinations, April - 2018**
**COMPUTER ORGANIZATION**
**(Computer Science and Engineering)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

**PART- A**

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | Define different types of interrupts. | [2] |
| b) | What are the different status bit conditions? | [3] |
| c) | Differentiate Isolated Vs Memory mapped I/O. | [2] |
| d) | Draw the block and timing diagram of handshaking. | [3] |
| e) | Define hit ratio. | [2] |
| f) | Define mapping and what the types of mapping techniques are. | [3] |
| g) | What is the function of READY pin of 8086 processor? | [2] |
| h) | What is the purpose of segmentation? | [3] |
| i) | Explain the Shift instructions in 8086. | [2] |
| j) | What is the difference between SUB instruction and CMP instruction? | [3] |

**PART-B**

**(50 Marks)**

2.a) What is addressing mode? Explain different addressing modes with numeric example.
b) Explain 3-address, 2-address, 1-address, zero-address instruction.
X= (A+B)*(C+D) [5+5]

**OR**

3.a) What are the different types of CPU organization .Explain each organization in detail with an example.
b) Define subroutine call and return. Explain different conditional branch instructions with tested condition and mnemonics. [5+5]

4.a) What is DMA? Explain the block diagram of DMA controller.
b) Differentiate programmed I/O and interrupt-initiated I/O mode of transfer. [5+5]

**OR**

5.a) Explain the block diagram of DMA transfer in a computer system.
b) Explain IOP-CPU-IOP communication in detail. [5+5]

6.a) Differentiate different types of ROMs.
b) A block set associated cache consists of a total of 64 blocks divided into 4-block sets. The main memory contains 4096 blocks, each containing of 128 words.
i) How many bits are there in main memory address?
ii) How many bits are there in each of tag, set and word fields? [5+5]

**OR**

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7.a) Define virtual memory. Explain how a virtual address is mapped to a physical address using mapping table with an example.

b) Differentiate static RAM and dynamic RAM chips. [5+5]

8.a) Explain the pins of 8086 in Maximum Mode.

b) Explain the process of 20 bit address generation using 16 bit registers in 8086 (Segmented Memory). [5+5]

**OR**

9.a) Explain the following Addressing Modes with suitable examples
i) Direct ii) Immediate and iii) Based-Indexed.

b) How is pipelining implemented in 8086? [5+5]

10.a) Explain following instructions in 8086 with suitable examples
i) AAA ii) XOR iii) STD

b) Differentiate between RET and IRET instruction in 8086. [5+5]

**OR**

11.a) Write an assembly language program in 8086 to sort in ascending order.

b) Explain about any two string manipulation instructions in 8086 in detail. [5+5]

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